each index i. We call X_i the *ith coordinate set.* When there can be no misunderstanding about the index set, the symbol $P_{i\in I}\, X_i$ is often abbreviated to $P_i X_i$. The definition we have just given requires that each coordinate set be non-empty before the product can be formed. It will be useful if we extend this definition slightly by agreeing that if any of the X_i's are empty, then $P_i X_i$ is also empty.

This approach to the idea of the product of a class of sets, by means of functions defined on the index set, is useful mainly in giving the definition. In practice, it is much more convenient to use the subscript notation x_i instead of the function notation $x(i)$. We then interpret the product $P_i X_i$ as made up of elements x, each of which is specified by the exhibited array $\{x_i\}$ of its values in the respective coordinate sets X_i. We call x_i the *ith coordinate* of the element $x = \{x_i\}$.

The mapping p_i of the product $P_i X_i$ onto its ith coordinate set X_i which is defined by $p_i(x) = x_i$—that is, the mapping whose value at an arbitrary element of the product is the ith coordinate of that element —is called the *projection* onto the ith coordinate set. The projection p_i selects the ith coordinate of each element in its domain. There is clearly one projection for each element of the index set I, and the set of all projections plays an important role in the general theory of topological spaces.

Problems

1. The graph of a mapping $f: X \to Y$ is a subset of the product $X \times Y$. What properties characterize the graphs of mappings among all subsets of $X \times Y$?
2. Let X and Y be non-empty sets. If A_1 and A_2 are subsets of X, and B_1 and B_2 subsets of Y, show the following:

$$(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2);$$
$$(A_1 \times B_1) - (A_2 \times B_2) = (A_1 - A_2) \times (B_1 - B_2) \cup (A_1 \cap A_2) \times (B_1 - B_2) \cup (A_1 - A_2) \times (B_1 \cap B_2).$$

3. Let X and Y be non-empty sets, and let $\mathbf{A}$ and $\mathbf{B}$ be rings of subsets of X and Y, respectively. Show that the class of all finite unions of sets of the form $A \times B$ with $A \in \mathbf{A}$ and $B \in \mathbf{B}$ is a ring of subsets of $X \times Y$.

5. PARTITIONS AND EQUIVALENCE RELATIONS

In the first part of this section we consider a non-empty set X, and we study decompositions of X into non-empty subsets which fill it out

and have no elements in common with one another. We give special attention to the tools (equivalence relations) which are normally used to generate such decompositions.

A *partition* of X is a disjoint class $\{X_i\}$ of non-empty subsets of X whose union is the full set X itself. The X_i's are called the *partition sets.* Expressed somewhat differently, a partition of X is the result of splitting it, or subdividing it, into non-empty subsets in such a way that each element of X belongs to one and only one of the given subsets.

If X is the set $\{1, 2, 3, 4, 5\}$, then $\{1, 3, 5\}$, $\{2, 4\}$ and $\{1, 2, 3\}$, $\{4, 5\}$ are two different partitions of X. If X is the set R of all real numbers, then we can partition X into the set of all rationals and the set of all irrationals, or into the infinitely many closed-open intervals of the form $[n, n + 1)$ where n is an integer. If X is the set of all points in the coordinate plane, then we can partition X in such a way that each partition set consists of all points with the same x coordinate (vertical lines), or so that each partition set consists of all points with the same y coordinate (horizontal lines).

Other partitions of each of these sets will readily occur to the reader. In general, there are many different ways in which any given set can be partitioned. These manufactured examples are admittedly rather uninspiring and serve only to make our ideas more concrete. Later in this section we consider some others which are more germane to our present purposes.

A *binary relation* in the set X is a mathematical symbol or verbal phrase, which we denote by R in this paragraph, such that for each ordered pair (x,y) of elements of X the statement $x \mathrm{R} y$ is meaningful, in the sense that it can be classified definitely as true or false. For such a binary relation, $x \mathrm{R} y$ symbolizes the assertion that x *is* related by R to y, and $x \not\mathrm{R} y$ the negation of this, namely, the assertion that x *is not* related by R to y. Many examples of binary relations can be given, some familiar and others less so, some mathematical and others not. For instance, if X is the set of all integers and R is interpreted to mean "is less than," which of course is usually denoted by the symbol $<$, then we clearly have $4 < 7$ and $5 \not< 2$. We have been speaking of binary relations, which are so named because they apply only to ordered pairs of elements, rather than to ordered triples, etc. In our work we drop the qualifying adjective and speak simply of a *relation* in X, since we shall have occasion to consider only relations of this kind.[1]

We now assume that a partition of our non-empty set X is given,

[1] Some writers prefer to regard a relation R in X as a subset R of $X \times X$. From this point of view, $x \mathrm{R} y$ and $x \not\mathrm{R} y$ are simply equivalent ways of writing $(x,y) \in \mathrm{R}$ and $(x,y) \notin \mathrm{R}$. This definition has the advantage of being more tangible than ours, and the disadvantage that few people really think of a relation in this way.

and we associate with this partition a relation in X. This relation is defined in the following way: we say that x is *equivalent* to y and write this $x \sim y$ (the symbol $\sim$ is pronounced "wiggle"), if x and y belong to the same partition set. It is obvious that the relation $\sim$ has the following properties:

(1) $x \sim x$ for every x (*reflexivity*);

(2) $x \sim y \Rightarrow y \sim x$ (*symmetry*);

(3) $x \sim y$ and $y \sim z \Rightarrow x \sim z$ (*transitivity*).

This particular relation in X arose in a special way, in connection with a given partition of X, and its properties are immediate consequences of its definition. Any relation whatever in X which possesses these three properties is called an *equivalence relation* in X.

We have just seen that each partition of X has associated with it a natural equivalence relation in X. We now reverse the situation and show that a given equivalence relation in X determines a natural partition of X.

Let $\sim$ be an equivalence relation in X; that is, assume that it is reflexive, symmetric, and transitive in the sense described above. If x is an element of X, the subset of X defined by $[x] = \{y : y \sim x\}$ is called the *equivalence set* of x. The equivalence set of x is thus the set of all elements which are equivalent to x. We show that the class of all distinct equivalence sets forms a partition of X. By reflexivity, $x \in [x]$ for each element x in X, so each equivalence set is non-empty and their union is X. It remains to be shown that any two equivalence sets $[x_1]$ and $[x_2]$ are either disjoint or identical. We prove this by showing that if $[x_1]$ and $[x_2]$ are not disjoint, then they must be identical. Suppose that $[x_1]$ and $[x_2]$ are not disjoint; that is, suppose that they have a common element z. Since z belongs to both equivalence sets, $z \sim x_1$ and $z \sim x_2$, and by symmetry, $x_1 \sim z$. Let y be any element of $[x_1]$, so that $y \sim x_1$. Since $y \sim x_1$ and $x_1 \sim z$, transitivity shows that $y \sim z$. By another application of transitivity, $y \sim z$ and $z \sim x_2$ imply that $y \sim x_2$, so that y is in $[x_2]$. Since y was chosen arbitrarily in $[x_1]$, we see by this that $[x_1] \subseteq [x_2]$. The same reasoning shows that $[x_2] \subseteq [x_1]$, and from this we conclude (see the last paragraph of Sec. 1) that $[x_1] = [x_2]$.

The above discussion demonstrates that there is no real distinction (other than a difference in language) between partitions of a set and equivalence relations in the set. If we start with a partition, we get an equivalence relation by regarding elements as equivalent if they belong to the same partition set, and if we start with an equivalence relation, we get a partition by grouping together into subsets all elements which are equivalent to one another. We have here a single mathematical idea, which we have been considering from two different points of view, and the approach we choose in any particular application depends entirely

on our own convenience. In practice, it is almost invariably the case that we use equivalence relations (which are usually easy to define) to obtain partitions (which are sometimes difficult to describe fully).

We now turn to several of the more important simple examples of equivalence relations.

Let I be the set of all integers. If a and b are elements of this set, we write $a = b$ (and say that a *equals* b) if a and b are the same integer. Thus $2 + 3 = 5$ means that the expressions on the left and right are simply different ways of writing the same integer. It is apparent that $=$ used in this sense is an equivalence relation in the set I:

(1) $a = a$ for every a;

(2) $a = b \Rightarrow b = a$;

(3) $a = b$ and $b = c \Rightarrow a = c$.

Clearly, each equivalence set consists of precisely one integer.

Another familiar example is the relation of equality commonly used for fractions. We remind the reader that, strictly speaking, a fraction is merely a symbol of the form a/b, where a and b are integers and b is not zero. The fractions $\frac{2}{3}$ and $\frac{4}{6}$ are obviously not identical, but nevertheless we consider them to be equal. In general, we say that two fractions a/b and c/d are *equal*, written $a/b = c/d$, if ad and bc are equal as integers in the usual sense (see the above paragraph). We leave it to the reader to show that this is an equivalence relation in the set of all fractions. An equivalence set of fractions is what we call a *rational number*. Everyday usage ignores the distinction between fractions and rational numbers, but it is important to recognize that from the strict point of view it is the rational numbers (and not the fractions) which form part of the real number system.

Our final example has a deeper significance, for it provides us with the basic tool for our work of the next two sections.

For the remainder of this section we consider a relation between pairs of non-empty sets, and each set mentioned (whether we say so explicitly or not) is assumed to be non-empty. If X and Y are two sets, we say that X is *numerically equivalent* to Y if there exists a one-to-one correspondence between X and Y, i.e., if there exists a one-to-one mapping of X onto Y. This relation is reflexive, since the identity mapping $i_X : X \to X$ is one-to-one onto; it is symmetric, since if $f : X \to Y$ is one-to-one onto, then its inverse mapping $f^{-1} : Y \to X$ is also one-to-one onto; and it is transitive, since if $f : X \to Y$ and $g : Y \to Z$ are one-to-one onto, then $gf : X \to Z$ is also one-to-one onto. Numerical equivalence has all the properties of an equivalence relation, and if we consider it as an equivalence relation in the class of all non-empty subsets of some universal set U, it groups together into equivalence sets all those subsets of U which have the *same number of elements*. After we state and prove the

following very useful but rather technical theorem, we shall continue in Secs. 6 and 7 with an exploration of the implications of these ideas.

The theorem we have in mind—the *Schroeder-Bernstein theorem*—is the following: *if X and Y are two sets each of which is numerically equivalent to a subset of the other, then all of X is numerically equivalent to all of Y.* There are several proofs of this classic theorem, some of which are quite difficult. The very elegant proof we give is essentially due to Birkhoff and MacLane.

Now for the proof. We assume that $f: X \to Y$ is a one-to-one mapping of X into Y, and that $g: Y \to X$ is a one-to-one mapping of Y into X. Our task is to produce a mapping $F: X \to Y$ which is one-to-one *onto*. We may assume that neither f nor g is onto, since if f is, we can define F to be f, and if g is, we can define F to be g^{-1}. Since both f and g are one-to-one, it is permissible to use the mappings f^{-1} and g^{-1} as long as we clearly understand that f^{-1} is defined only on $f(X)$ and g^{-1} only on $g(Y)$. We obtain the mapping F by splitting both X and Y into subsets which we characterize in terms of the ancestry of their elements. Let x be an element of X. We apply g^{-1} to it (if we can) to get the element $g^{-1}(x)$ in Y. If $g^{-1}(x)$ exists, we call it the first ancestor of x. The element x itself we call the zeroth ancestor of x. We now apply f^{-1} to $g^{-1}(x)$ if we can, and if $(f^{-1}g^{-1})(x)$ exists, we call it the second ancestor of x. We now apply g^{-1} to $(f^{-1}g^{-1})(x)$ if we can, and if $(g^{-1}f^{-1}g^{-1})(x)$ exists, we call it the third ancestor of x. As we continue this process of tracing back the ancestry of x, it becomes apparent that there are three possibilities. (1) x has infinitely many ancestors. We denote by X_i the subset of X which consists of all elements with infinitely many ancestors. (2) x has an even number of ancestors; this means that x has a last ancestor (that is, one which itself has no first ancestor) in X. We denote by X_e the subset of X consisting of all elements with an even number of ancestors. (3) x has an odd number of ancestors; this means that x has a last ancestor in Y. We denote by X_o the subset of X which consists of all elements with an odd number of ancestors. The three sets X_i, X_e, X_o form a disjoint class whose union is X. We decompose Y in just the same way into three subsets Y_i, Y_e, Y_o. It is easy to see that f maps X_i onto Y_i and X_e onto Y_o, and that g^{-1} maps X_o onto Y_e; and we complete the proof by defining F in the following piecemeal manner:

$$F(x) = \begin{cases} f(x) & \text{if } x \in X_i \cup X_e, \\ g^{-1}(x) & \text{if } x \in X_o. \end{cases}$$

We attempt to illustrate these ideas in Fig. 12. Here we present two replicas of the situation: on the left, X and Y are represented by the vertical lines, and f and g by the lines slanting down to the right and

left; and on the right, we schematically trace the ancestry of three elements in X, of which x_1 has no first ancestor, x_2 has a first and second ancestor, and x_3 has a first, second, and third ancestor.

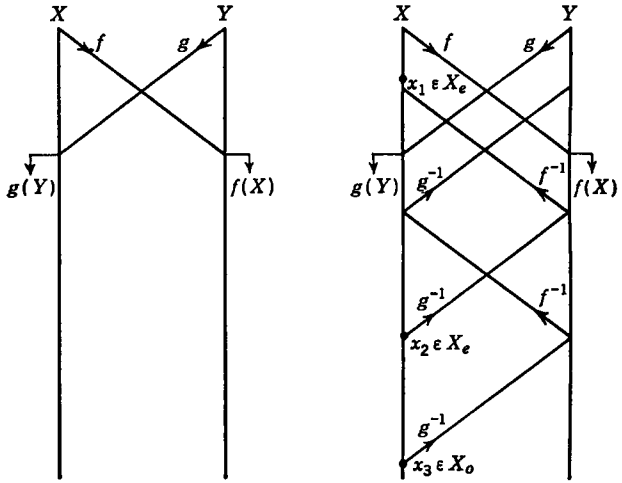


Fig. 12. The proof of the Schroeder-Bernstein theorem.

The Schroeder-Bernstein theorem has great theoretical and practical significance. Its main value for us lies in its role as a tool by means of which we can prove numerical equivalence with a minimum of effort for many specific sets. We put it to work in Sec. 7.

Problems

1. Let f: $X \to Y$ be an arbitrary mapping. Define a relation in X as follows: $x_1 \sim x_2$ means that $f(x_1) = f(x_2)$. Show that this is an equivalence relation and describe the equivalence sets.
2. In the set R of all real numbers, let $x \sim y$ mean that $x - y$ is an integer. Show that this is an equivalence relation and describe the equivalence sets.
3. Let I be the set of all integers, and let m be a fixed positive integer. Two integers a and b are said to be *congruent* modulo m—symbolized by $a \equiv b \pmod{m}$—if $a - b$ is exactly divisible by m, i.e., if $a - b$ is an integral multiple of m. Show that this is an equivalence relation, describe the equivalence sets, and state the number of distinct equivalence sets.
4. Decide which ones of the three properties of reflexivity, symmetry, and transitivity are true for each of the following relations in the set

of all positive integers: $m \leq n$, $m < n$, m divides n. Are any of these equivalence relations?

5. Give an example of a relation which is (*a*) reflexive but not symmetric or transitive; (*b*) symmetric but not reflexive or transitive; (*c*) transitive but not reflexive or symmetric; (*d*) reflexive and symmetric but not transitive; (*e*) reflexive and transitive but not symmetric; (*f*) symmetric and transitive but not reflexive.
6. Let X be a non-empty set and $\sim$ a relation in X. The following purports to be a proof of the statement that if this relation is symmetric and transitive, then it is necessarily reflexive: $x \sim y \Rightarrow y \sim x$; $x \sim y$ and $y \sim x \Rightarrow x \sim x$; therefore $x \sim x$ for every x. In view of Problem 5*f*, this cannot be a valid proof. What is the flaw in the reasoning?
7. Let X be a non-empty set. A relation $\sim$ in X is called *circular* if $x \sim y$ and $y \sim z \Rightarrow z \sim x$, and *triangular* if $x \sim y$ and $x \sim z \Rightarrow y \sim z$. Prove that a relation in X is an equivalence relation $\Leftrightarrow$ it is reflexive and circular $\Leftrightarrow$ it is reflexive and triangular.

6. COUNTABLE SETS

The subject of this section and the next—*infinite cardinal numbers*—lies at the very foundation of modern mathematics. It is a vital instrument in the day-to-day work of many mathematicians, and we shall make extensive use of it ourselves. This theory, which was created by the German mathematician Cantor, also has great aesthetic appeal, for it begins with ideas of extreme simplicity and develops through natural stages into an elaborate and beautiful structure of thought. In the course of our discussion we shall answer questions which no one before Cantor's time thought to ask, and we shall ask a question which no one can answer to this day.

Without further ado, we can say that *cardinal numbers* are those used in counting, such as the positive integers (or natural numbers) 1, 2, 3, . . . familiar to us all. But there is much more to the story than this.

The act of counting is undoubtedly one of the oldest of human activities. Men probably learned to count in a crude way at about the same time as they began to develop articulate speech. The earliest men who lived in communities and domesticated animals must have found it necessary to record the number of goats in the village herd by means of a pile of stones or some similar device. If the herd was counted in each night by removing one stone from the pile for each goat accounted for, then stones left over would have indicated strays, and herdsmen would have gone out to search for them. Names for numbers and symbols for

them, like our 1, 2, 3, . . . , would have been superfluous. The simple and yet profound idea of a one-to-one correspondence between the stones and the goats would have fully met the needs of the situation.

In a manner of speaking, we ourselves use the infinite set

$$N = \{1, 2, 3, \ldots\}$$

of all positive integers as a "pile of stones." We carry this set around with us as part of our intellectual equipment. Whenever we want to count a set, say, a stack of dollar bills, we start through the set N and tally off one bill against each positive integer as we come to it. The last number we reach, corresponding to the last bill, is what we call the number of bills in the stack. If this last number happens to be 10, then "10" is our symbol for the number of bills in the stack, as it also is for the number of our fingers, and for the number of our toes, and for the number of elements in any set which can be put into one-to-one correspondence with the finite set $\{1, 2, \ldots, 10\}$. Our procedure is slightly more sophisticated than that of the primitive savage. We have the symbols 1, 2, 3, . . . for the numbers which arise in counting; we can record them for future use, and communicate them to other people, and manipulate them by the operations of arithmetic. But the underlying idea, that of the one-to-one correspondence, remains the same for us as it probably was for him.

The positive integers are adequate for the purpose of counting any non-empty finite set, and since outside of mathematics all sets appear to be of this kind, they suffice for all non-mathematical counting. But in the world of mathematics we are obliged to consider many infinite sets, such as the set of all positive integers itself, the set of all integers, the set of all rational numbers, the set of all real numbers, the set of all points in a plane, and so on. It is often important to be able to count such sets, and it was Cantor's idea to do this, and to develop a theory of infinite cardinal numbers, by means of one-to-one correspondences.

In comparing the sizes of two sets, the basic concept is that of numerical equivalence as defined in the previous section. We recall that two non-empty sets X and Y are said to be numerically equivalent if there exists a one-to-one mapping of one onto the other, or—and this amounts to the same thing—if there can be found a one-to-one correspondence between them. To say that two non-empty finite sets are numerically equivalent is of course to say that they have the *same number of elements* in the ordinary sense. If we count one of them, we simply establish a one-to-one correspondence between its elements and a set of positive integers of the form $\{1, 2, \ldots, n\}$, and we then say that n is the *number of elements possessed by both*, or the *cardinal number of both*. The positive integers are the *finite cardinal numbers*. We encounter

many surprises as we follow Cantor and consider numerical equivalence for infinite sets.

The set $N = \{1, 2, 3, \ldots\}$ of all positive integers is obviously "larger" than the set $\{2, 4, 6, \ldots\}$ of all even positive integers, for it contains this set as a proper subset. It appears on the surface that N has "more" elements. But it is very important to avoid jumping to conclusions when dealing with infinite sets, and we must remember that our criterion in these matters is whether there exists a one-to-one correspondence between the sets (not whether one set is or is not a proper subset of the other). As a matter of fact, the pairing

$$\begin{array}{l} 1, 2, 3, \ldots, n, \ldots \\ 2, 4, 6, \ldots, 2n, \ldots \end{array}$$

serves to establish a one-to-one correspondence between these sets, in which each positive integer in the upper row is matched with the even positive integer (its double) directly below it, and these two sets must therefore be regarded as having the *same number of elements.* This is a very remarkable circumstance, for it seems to contradict our intuition and yet is based only on solid common sense. We shall see below, in Problems 6 and 7-4, that every infinite set is numerically equivalent to a proper subset of itself. Since this property is clearly not possessed by any finite set, some writers even use it as the definition of an infinite set.

In much the same way as above, we can show that N is numerically equivalent to the set of *all* even integers:

$$\begin{array}{l} 1, 2, \quad 3, 4, \quad 5, 6, \quad 7, \ldots \\ 0, 2, -2, 4, -4, 6, -6, \ldots \end{array}$$

Here our device is to start with 0 and follow each even positive integer as we come to it by its negative. Similarly, N is numerically equivalent to the set of all integers:

$$\begin{array}{l} 1, 2, \quad 3, 4, \quad 5, 6, \quad 7, \ldots \\ 0, 1, -1, 2, -2, 3, -3, \ldots \end{array}$$

It is of considerable historical interest to note that Galileo observed in the early seventeenth century that there are precisely as many perfect squares (1, 4, 9, 16, 25, etc.) among the positive integers as there are positive integers altogether. This is clear from the pairing

$$\begin{array}{l} 1, \ 2, \ 3, \ 4, \ 5, \ldots \\ 1^2, 2^2, 3^2, 4^2, 5^2, \ldots \end{array}$$

It struck him as very strange that this should be true, considering how

sparsely strewn the squares are among all the positive integers. But the time appears not to have been ripe for the exploration of this phenomenon, or perhaps he had other things on his mind; in any case, he did not follow up his idea.

These examples should make it clear that all that is really necessary in showing that an infinite set X is numerically equivalent to N is that we be able to list the elements of X, with a first, a second, a third, and so on, in such a way that it is completely exhausted by this counting off of its elements. It is for this reason that any infinite set which is numerically equivalent to N is said to be *countably infinite*. We say that a set is *countable* if it is non-empty and finite (in which case it can obviously be counted) or if it is countably infinite.

One of Cantor's earliest discoveries in his study of infinite sets was that the set of all positive rational numbers (which is very large: it contains N and a great many other numbers besides) is actually countable. We cannot list the positive rational numbers in order of size, as we can the positive integers, beginning with the smallest, then the next smallest, and so on, for there is no smallest, and between any two there are infinitely many others. We must find some other way of counting them, and following Cantor, we arrange them not in order of size, but according to the size of the sum of the numerator and denominator. We begin with all positive rationals whose numerator and denominator add up to 2: there is only one, $\frac{1}{1} = 1$. Next we list (with increasing numerators) all those for which this sum is 3: $\frac{1}{2}$, $\frac{2}{1} = 2$. Next, all those for which this sum is 4: $\frac{1}{3}$, $\frac{2}{2} = 1$, $\frac{3}{1} = 3$. Next, all those for which this sum is 5: $\frac{1}{4}$, $\frac{2}{3}$, $\frac{3}{2}$, $\frac{4}{1} = 4$. Next, all those for which this sum is 6: $\frac{1}{5}$, $\frac{2}{4} = \frac{1}{2}$, $\frac{3}{3} = 1$, $\frac{4}{2} = 2$, $\frac{5}{1} = 5$. And so on. If we now list all these together from the beginning, omitting those already listed when we come to them, we get a sequence

$$1, \tfrac{1}{2}, 2, \tfrac{1}{3}, 3, \tfrac{1}{4}, \tfrac{2}{3}, \tfrac{3}{2}, 4, \tfrac{1}{5}, 5, \ldots$$

which contains each positive rational number once and only once. Figure 13 gives a schematic representation of this manner of listing the positive rationals. In this figure the first row contains all positive rationals with numerator 1, the second all with numerator 2, etc.; and the first column contains all with denominator 1, the second all with denominator 2, and so on. Our listing amounts to traversing this array of numbers as the arrows indicate, where of course all those numbers already encountered are left out.

It's high time that we christened the infinite cardinal number we've been discussing, and for this purpose we use the first letter of the Hebrew alphabet ($\aleph$, pronounced "aleph") with 0 as a subscript. We say that $\aleph_0$ is the number of elements in any countably infinite set. Our

complete list of cardinal numbers so far is

$$1, 2, 3, \ldots, \aleph_0.$$

We expand this list in the next section.

Suppose now that m and n are two cardinal numbers (finite or infinite). The statement that m is *less than* n (written $m < n$) is defined to mean the following: if X and Y are sets with m and n elements, then

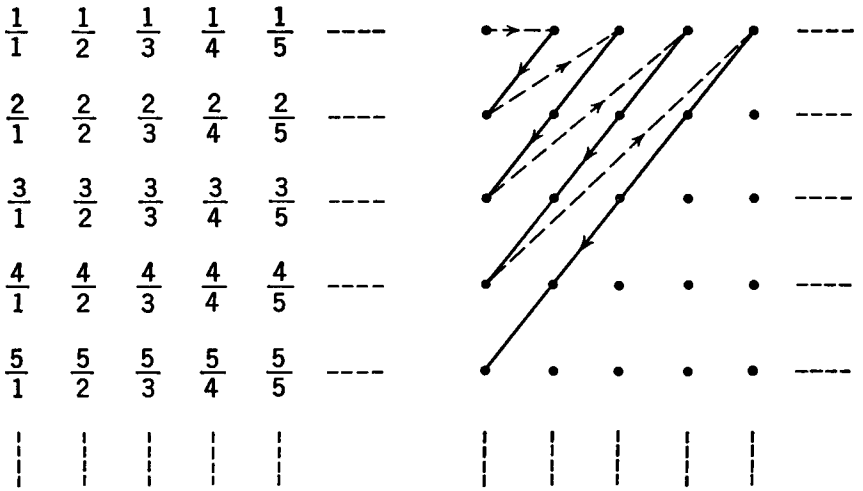


Fig. 13. A listing of the positive rationals.

(1) there exists a one-to-one mapping of X into Y, and (2) there does not exist a one-to-one mapping of X onto Y. Using this concept, it is easy to relate our cardinal numbers to one another by means of

$$1 < 2 < 3 < \cdots < \aleph_0.$$

With respect to the finite cardinal numbers, this ordering corresponds to their usual ordering as real numbers.

Problems

1. Prove that the set of all rational numbers (positive, negative, and zero) is countable. (*Hint:* see our method of showing that the set of all integers is countable.)
2. Use the idea behind Fig. 13 to prove that if $\{X_i\}$ is a countable class of countable sets, then $\cup_i X_i$ is also countable. We usually express this by saying that *any countable union of countable sets is countable.*

3. Prove that the set of all rational points in the coordinate plane R^2 (i.e., all points whose coordinates are both rational) is countable.
4. Prove that if X_1 and X_2 are countable, then $X_1 \times X_2$ is also countable.
5. Prove that if $X_1, X_2, \ldots, X_n$ are countable, where n is any positive integer, then $X_1 \times X_2 \times \cdots \times X_n$ is also countable.
6. Prove that every countably infinite set is numerically equivalent to a proper subset of itself.
7. Prove that any non-empty subset of a countable set is countable.
8. Let X and Y be non-empty sets, and f a mapping of X onto Y. If X is countable, prove that Y is also countable.

7. UNCOUNTABLE SETS

All the infinite sets we considered in the previous section were countable, so it might appear at this stage that *every* infinite set is countable. If this were true, if the end result of the analysis of infinite sets were that they are all numerically equivalent to one another, then Cantor's theory would be relatively trivial. But this is not the case, for Cantor discovered that the infinite set R of all real numbers is *not* countable—or, as we phrase it, R is *uncountable* or *uncountably infinite*. Since we customarily identify the elements of R with the points of the real line (see Sec. 4), this amounts to the assertion that the set of *all* points on the real line represents a "higher type of infinity" than that of only the integral points or only the rational points.

Cantor's proof of this is very ingenious, but it is actually quite simple. In outline the procedure is as follows: we assume that all the real numbers (in decimal form) can be listed, and in fact have been listed; then we produce a real number which cannot be in this list—thus contradicting our initial assumption that a complete listing is possible. In representing real numbers by decimals, we use the scheme of decimal expansion in which infinite chains of 9's are avoided; for instance, we write $\frac{1}{2}$ as .5000 . . . and not as .4999 . . . . In this way we guarantee that each real number has one and only one decimal representation. Suppose now that we can list all the real numbers, and that they have been listed in a column like the one below (where we use particular numbers for the purpose of illustration).

1st number	$13 + .712983 \ldots$
2nd number	$-4 + .913572 \ldots$
3rd number	$0 + .843265 \ldots$
. . . . .	. . . . .

Since it is impossible actually to write down this infinite list of decimals, our assumption that all the real numbers can be listed in this way means that we assume that we have available some general rule according to which the list is constructed, similar to that used for listing the positive rationals, and that every conceivable real number occurs somewhere in this list. We now demonstrate that this assumption is false by exhibiting a decimal $.a_1a_2a_3 \ldots$ which is constructed in such a way that it is not in the list. We choose a_1 to be 1 unless the first digit after the decimal point of the first number in our list is 1, in which case we choose a_1 to be 2. Clearly, our new decimal will differ from the first number in our list regardless of how we choose its remaining digits. Next, we choose a_2 to be 1 unless the second digit after the decimal point of the second number in our list is 1, in which case we choose a_2 to be 2. Just as above, our new decimal will necessarily differ from the second number in our list. We continue building up the decimal $.a_1a_2a_3 \ldots$ in this way, and since the process can be continued indefinitely, it defines a real number in decimal form (.121 . . . in the case of our illustrative example) which is different from each number in our list. This contradicts our assumption that we can list all the real numbers and completes our proof of the fact that the set R of all real numbers is uncountable.

We have seen (in Problem 6-1) that the set of all rational points on the real line is countable, and we have just proved that the set of *all* points on the real line is uncountable. We conclude at once from this that irrational points on the real line (i.e., irrational numbers) must exist. In fact, it is very easy to see by means of Problem 6-2 that the set of all irrational numbers is uncountably infinite. To vary slightly a striking metaphor coined by E. T. Bell, the rational numbers are spotted along the real line like stars against a black sky, and the dense blackness of the background is the firmament of the irrationals. The reader is probably familiar with a proof of the fact that the square root of 2 is irrational. This proof demonstrates the existence of irrational numbers by exhibiting a specimen. Our remarks, on the other hand, do not show that this or that particular number is irrational; they merely show that such numbers must exist, and moreover must exist in overwhelming abundance.

If the reader supposes that the set of all points on the real line R is uncountable because R is infinitely long, then we can disillusion him by the following argument, which shows that any open interval on R, no matter how short it may be, has precisely as many points as R itself. Let a and b be any two real numbers with $a < b$, and consider the open interval (a,b). Figure 14 shows how to establish a one-to-one correspondence between the points P of (a,b) and the points P' of R: we bend (a,b) into a semicircle; we rest this semicircle tangentially on the

real line R as shown in the figure; and we link P and P' by projecting from its center. If formulas are preferred over geometric reasoning of this kind, we observe that $y = a + (b - a)x$ is a numerical equivalence between real numbers $x \in (0,1)$ and $y \in (a,b)$, and that $z = \tan \pi(x - \frac{1}{2})$ is another numerical equivalence between $(0,1)$ and all of R. It now follows that (a,b) and R are numerically equivalent to one another.

We are now in a position to show that any subset X of the real line R which contains an open interval I is numerically equivalent to R, no matter how complicated the structure of X may be. The proof of this fact is very simple, and it uses only the Schroeder-Bernstein theorem and our above result that I is numerically equivalent to R. The argument can be given in two sentences. Since X is numerically equivalent to itself, it is obviously numerically equivalent to a subset of R; and R is numerically equivalent to a subset of X, namely, to I. It is now a direct consequence of the Schroeder-Bernstein theorem that X and R are numerically equivalent to one another. We point out that all numerical equivalences up to this point have been established by actually exhibiting one-to-one correspondences between the sets concerned. In the present situation, however, it is not feasible to do this, for very little has been assumed about the specific nature of the set X. Without the help of the Schroeder-Bernstein theorem it would be very difficult to prove theorems of this type.

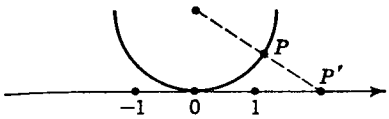


Fig. 14. A one-to-one correspondence between an open interval and the real line.

We give another interesting application of the Schroeder-Bernstein theorem. Consider the coordinate plane R^2 and the subset X of R^2 defined by $X = \{(x,y) : 0 \leq x < 1 \text{ and } 0 \leq y < 1\}$. We show that X is numerically equivalent to the closed-open interval

$$I = \{(x,y) : 0 \leq x < 1 \text{ and } y = 0\}$$

which forms its base (see Fig. 15). Since I is numerically equivalent to a subset of X, namely, to I itself, our conclusion will follow at once from the Schroeder-Bernstein theorem if we can establish a one-to-one mapping of X into I. This we now do. Let (x,y) be an arbitrary point of X. Each of the coordinates x and y has a unique decimal expansion which does not end in an infinite chain of 9's. We form another decimal z from these by alternating their digits; for example, if $x = .327 \ldots$ and $y = .614 \ldots$, then $z = .362174 \ldots$. We now identify z (which cannot end in an infinite chain of 9's) with a point of I. This gives the required one-to-one mapping of X into I and yields the somewhat

startling result that there are no more points inside a square than there are on one of its sides.

In Sec. 6 we introduced the symbol $\aleph_0$ for the number of elements in any countably infinite set. At the beginning of this section we proved that the set R of all real numbers (or of all points on the real line) is uncountably infinite. We now introduce the symbol c (called the *cardinal number of the continuum*) for the number of elements in R. c is the cardinal number of R and of any set which is numerically equivalent to R. In the above three paragraphs we have demonstrated that c is the cardinal number of any open interval, of any subset of R which contains an open interval, and of the subset X of the coordinate plane which is illustrated in Fig. 15. Our list of cardinal numbers has now grown to

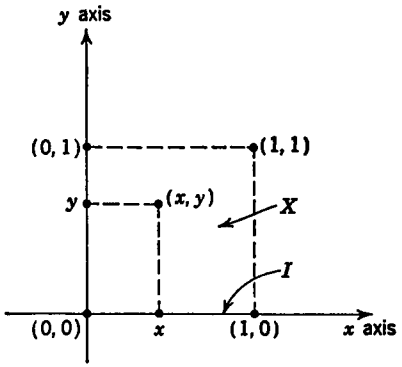


Fig. 15

$$1, 2, 3, \ldots, \aleph_0, c,$$

and they are related to each other by

$$1 < 2 < 3 < \cdots < \aleph_0 < c.$$

At this point we encounter one of the most famous unsolved problems of mathematics. Is there a cardinal number greater than $\aleph_0$ and less than c? No one knows the answer to this question. Cantor himself thought that there is no such number, or in other words, that c is the next infinite cardinal number greater than $\aleph_0$, and his guess has come to be known as *Cantor's continuum hypothesis.* The continuum hypothesis can also be expressed by the assertion that every uncountable set of real numbers has c as its cardinal number.[1]

There is another question which arises naturally at this stage, and this one we are fortunately able to answer. Are there any infinite cardinal numbers greater than c? Yes, there are; for example, the cardinal number of the class of all subsets of R. This answer depends on the following fact: if X is any non-empty set, then the cardinal number of X is less than the cardinal number of the class of all subsets of X.

We prove this statement as follows. In accordance with the definition given in the last paragraph of the previous section, we must show

[1] For further information about the continuum hypothesis, see Wilder [42, p. 125] and Gödel [12].

(1) that there exists a one-to-one mapping of X into the class of all its subsets, and (2) that there does not exist such a mapping of X onto this class. To prove (1), we have only to point to the mapping $x \to \{x\}$, which makes correspond to each element x that set $\{x\}$ which consists of the element x alone. We prove (2) indirectly. Let us assume that there does exist a one-to-one mapping f of X onto the class of all its subsets. We now deduce a contradiction from the assumed existence of such a mapping. Let A be the subset of X defined by $A = \{x : x \notin f(x)\}$. Since our mapping f is onto, there must exist an element a in X such that $f(a) = A$. Where is the element a? *If* a is in A, then by the definition of A we have $a \notin f(a)$, and since $f(a) = A$, $a \notin A$. This is a contradiction, so a cannot belong to A. But if a is not in A, then again by the definition of A we have $a \in f(a)$ or $a \in A$, which is another contradiction. The situation is impossible, so our assumption that such a mapping exists must be false.

This result guarantees that given any cardinal number, there always exists a greater one. If we start with a set $X_1 = \{1\}$ containing one element, then there are two subsets, the empty set $\emptyset$ and the set $\{1\}$ itself. If $X_2 = \{1,2\}$ is a set containing two elements, then there are four subsets: $\emptyset$, $\{1\}$, $\{2\}$, $\{1,2\}$. If $X_3 = \{1, 2, 3\}$ is a set containing three elements, then there are eight subsets: $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, $\{1, 2, 3\}$. In general, if X_n is a set with n elements, where n is any finite cardinal number, then X_n has 2^n subsets. If we now take n to be any infinite cardinal number, the above facts suggest that we *define* 2^n to be the number of subsets of any set with n elements. If n is the first infinite cardinal number, namely, $\aleph_0$, then it can be shown that

$$2^{\aleph_0} = c.$$

The simplest proof of this fact depends on the ideas developed in the following paragraph.

Consider the closed-open unit interval $[0,1)$ and a real number x in this set. Our concern is with the meaning of the *decimal*, *binary*, and *ternary expansions* of x. For the sake of clarity, let us take x to be $\frac{1}{4}$. How do we arrive at the decimal expansion of $\frac{1}{4}$? First, we split $[0,1)$ into the 10 closed-open intervals

$$[0,\tfrac{1}{10}), [\tfrac{1}{10},\tfrac{2}{10}), \ldots, [\tfrac{9}{10},1),$$

and we use the 10 digits 0, 1, . . . , 9 to number them in order. Our number $\frac{1}{4}$ belongs to exactly one of these intervals, namely, to $[\frac{2}{10},\frac{3}{10})$. We have labeled this interval with the digit 2, so 2 is the first digit after the decimal point in the decimal expansion of $\frac{1}{4}$:

$$\tfrac{1}{4} = .2 \ldots$$

Next, we split the interval $[2/10, 3/10)$ into the 10 closed-open intervals

$$[2/10, 21/100), [21/100, 22/100), \ldots, [29/100, 3/10),$$

and we use the 10 digits to number these in order. Our number $1/4$ belongs to $[25/100, 26/100)$, which is labeled with the digit 5, so 5 is the second number after the decimal point in the decimal expansion of $1/4$:

$$1/4 = .25 \ldots$$

If we continue this process exactly as we started it, we can obtain the decimal expansion of $1/4$ to as many places as we wish. As a matter of fact, if we do continue, we get 0 at each stage from this point on:

$$1/4 = .25000 \ldots$$

The reader should notice that there is no ambiguity in this system as we have explained it: contrary to customary usage, .24999 . . . is *not* to be regarded as another decimal expansion of $1/4$ which is "equivalent" to .25000 . . . . In this system, each real number x in $[0,1)$ has *one and only one* decimal expansion which cannot end in an infinite chain of 9's. There is nothing magical about the role of the number 10 in the above discussion. If at each stage we split our closed-open interval into two equal closed-open intervals, and if we use the two digits 0 and 1 to number them, we obtain the binary expansion of any real number x in $[0,1)$. The binary expansion of $1/4$ is easily seen to be .01000 . . . . The ternary expansion of x is found similarly: at each stage we split our closed-open interval into three equal closed-open intervals, and we use the three digits 0, 1, and 2 to number them. A moment's thought should convince the reader that the ternary expansion of $1/4$ is .020202 . . . . Just as (in our system) the decimal expansion of a number in $[0,1)$ cannot end in an infinite chain of 9's, so also its binary expansion cannot end in an infinite chain of 1's, and its ternary expansion cannot end in an infinite chain of 2's.

We now use this machinery to give a proof of the fact that

$$2^{\aleph_0} = c.$$

Consider the two sets $N = \{1, 2, 3, \ldots\}$ and $I = [0,1)$, the first with cardinal number $\aleph_0$ and the second with cardinal number c. If $\mathbf{N}$ denotes the class of all subsets of N, then by definition $\mathbf{N}$ has cardinal number $2^{\aleph_0}$. Our proof amounts to showing that there exists a one-to-one correspondence between $\mathbf{N}$ and I. We begin by establishing a one-to-one mapping f of $\mathbf{N}$ into I. If A is a subset of N, then $f(A)$ is that real number x in I whose decimal expansion $x = .d_1d_2d_3 \ldots$ is defined by the condition that d_n is 3 or 5 according as n is or is not in A. Any other two digits can be used here, as long as neither of them is 9. Next, we con-

struct a one-to-one mapping g of I into $\mathbf{N}$. If x is a real number in I, and if $x = .b_1b_2b_3 \ldots$ is its binary expansion (so that each b_n is either 0 or 1), then $g(x)$ is that subset A of N defined by $A = \{n : b_n = 1\}$. We conclude the proof with an appeal to the Schroeder-Bernstein theorem, which guarantees that under these conditions $\mathbf{N}$ and I are numerically equivalent to one another.

If we follow up the hint contained in the fact that $2^{\aleph_0} = c$, and successively form 2^c, 2^{2^c}, and so on, we get a chain of cardinal numbers

$$1 < 2 < 3 < \cdots < \aleph_0 < c < 2^c < 2^{2^c} < \cdots$$

in which there are infinitely many infinite cardinal numbers. Clearly, there is only one kind of countable infinity, symbolized by $\aleph_0$, and beyond this there is an infinite hierarchy of uncountable infinities which are all distinct from one another.

At this point we bring our discussion of these matters to a close. We have barely touched on Cantor's theory and have left entirely to one side, for instance, all questions relating to the addition and multiplication of infinite cardinal numbers and the rules of arithmetic which apply to these operations. We have developed these ideas, not for their own sake, but for the sake of their applications in algebra and topology, and our main purpose throughout the last two sections has been to give the reader some of the necessary insight into countable and uncountable sets and the distinction between them.[1]

Problems

1. Show geometrically that the set of all points in the coordinate plane R^2 is numerically equivalent to the subset X of R^2 illustrated in Fig. 15 and defined by $X = \{(x,y) : 0 \leq x < 1 \text{ and } 0 \leq y < 1\}$, and that therefore R^2 has cardinal number c. [*Hint:* rest an open hemispherical surface (= a hemispherical surface minus its boundary) tangentially on the center of X, project from various points on the line through its center and perpendicular to R^2, and use the Schroeder-Bernstein theorem.]
2. Show that the subset X of R^3 defined by

$$X = \{(x_1, x_2, x_3) : 0 \leq x_i < 1 \text{ for } i = 1, 2, 3\}$$

has cardinal number c.

[1] For the reader who wishes to learn something about the arithmetic of infinite cardinal numbers, we recommend Halmos [16, sec. 24], Kamke [24, chap. 2], Sierpinski [37, chaps. 7–10], or Fraenkel [9, chap. 2].

equipped with a concept of distance which is suitable for the treatment of convergent sequences in the set and continuous functions defined on the set. Our purpose in this chapter is to develop in a systematic manner the main elementary facts about metric spaces. These facts are important for their own sake, and also for the sake of the motivation they provide for our later work on topological spaces.

9. THE DEFINITION AND SOME EXAMPLES

Let X be a non-empty set. A *metric* on X is a real function d of ordered pairs of elements of X which satisfies the following three conditions:

(1) $d(x,y) \geq 0$, and $d(x,y) = 0 \Leftrightarrow x = y$;
(2) $d(x,y) = d(y,x)$ (*symmetry*);
(3) $d(x,y) \leq d(x,z) + d(z,y)$ (the *triangle inequality*).

The function d assigns to each pair (x,y) of elements of X a non-negative real number $d(x,y)$, which by symmetry does not depend on the order of the elements; $d(x,y)$ is called the *distance* between x and y. A *metric space* consists of two objects: a non-empty set X and a metric d on X. The elements of X are called the *points* of the metric space (X,d). Whenever it can be done without causing confusion, we denote the metric space (X,d) by the symbol X which is used for the underlying set of points. One should always keep in mind, however, that a metric space is not merely a non-empty set: it is a non-empty set together with a metric. It often happens that several different metrics can be defined on a single given non-empty set, and in this case distinct metrics make the set into distinct metric spaces.

There are many different kinds of metric spaces, some of which play very significant roles in geometry and analysis. Our first example is rather trivial, but it is often useful in showing that certain statements we might wish to make are not true. It also shows that every non-empty set can be regarded as a metric space.

Example 1. Let X be an arbitrary non-empty set, and define d by

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

The reader can easily see for himself that this definition yields a metric on X.

Our next two examples are the fundamental number systems of mathematics.

Example 2. Consider the real line R and the real function $|x|$ defined on R. Three elementary properties of this absolute value function are important for our purposes:

(i) $|x| \geq 0$, and $|x| = 0 \Leftrightarrow x = 0$;
(ii) $|-x| = |x|$;
(iii) $|x + y| \leq |x| + |y|$.

We now define a metric on R by

$$d(x,y) = |x - y|.$$

This is called the *usual metric* on R, and the real line, as a metric space, is always understood to have this as its metric. The fact that d actually is a metric follows from the three properties stated above. This is a piece of reasoning which occurs frequently in our work, so we give the details. By (i), $d(x,y) = |x - y|$ is a non-negative real number which equals $0 \Leftrightarrow x - y = 0 \Leftrightarrow x = y$. By (ii),

$$d(x,y) = |x - y| = |-(y - x)| = |y - x| = d(y,x).$$

And by (iii),

$$d(x,y) = |x - y| = |(x - z) + (z - y)| \leq |x - z| + |z - y| = d(x,z) + d(z,y).$$

Example 3. Consider the complex plane C. We mentioned C briefly in Sec. 4, and we described the sense in which it can be identified as a set with the coordinate plane R^2. We now give a somewhat fuller discussion. If z is a complex number, and if $z = a + ib$ where a and b are real numbers, then a and b are called the *real part* and the *imaginary part* of z and are denoted by $R(z)$ and $I(z)$. Two complex numbers are said to be *equal* if their real and imaginary parts are equal:

$$a + ib = c + id \Leftrightarrow a = c \text{ and } b = d.$$

We add (or subtract) two complex numbers by adding (or subtracting) their real and imaginary parts, and we multiply them by multiplying them out as in elementary algebra and replacing i^2 by -1 wherever it appears:

$$(a + ib) \pm (c + id) = (a \pm c) + i(b \pm d),$$

and

$$(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc).$$

Division is carried out in accordance with

$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \frac{(ac+bd)+i(bc-ad)}{c^2+d^2}$$
$$= \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2},$$

where $c^2 + d^2$ is required to be non-zero. If $z = a + ib$ is a complex number, then its *negative* $-z$ and its *conjugate* $\bar{z}$ are defined by

$$-z = (-a) + i(-b)$$

and $\bar{z} = a + i(-b)$, which are usually written more informally as $-z = -a - ib$ and $\bar{z} = a - ib$. It is easy to see that

$$R(z) = \frac{z+\bar{z}}{2} \quad \text{and} \quad I(z) = \frac{z-\bar{z}}{2i}.$$

The real line R is usually regarded as part of the complex plane:

$$R = \{z : I(z) = 0\} = \{z : \bar{z} = z\}.$$

Simple calculations show directly that

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \qquad \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2, \qquad \text{and} \qquad \bar{\bar{z}} = z.$$

The *origin*, or *zero*, is the complex number $0 = 0 + i0$. The ordinary distance from $z = a + ib$ to the origin is defined by

$$|z| = (a^2 + b^2)^{1/2}.$$

$|z|$ is called the *absolute value* of z, and it is easy to see that

$$|\bar{z}| = |z| \qquad \text{and} \qquad |z|^2 = z\bar{z}.$$

The *usual metric* on C is defined by

$$d(z_1, z_2) = |z_1 - z_2|.$$

Exactly as in Example 2, the fact that this is a metric is a consequence of the following properties of the real function $|z|$:

(i) $|z| \geq 0$, and $|z| = 0 \Leftrightarrow z = 0$;

(ii) $|-z| = |z|$;

(iii) $|z_1 + z_2| \leq |z_1| + |z_2|$.

Properties (i) and (ii) are obvious. Since $-z = (-1)z$, property (ii) is also a special case of the fact that

$$|z_1 z_2| = |z_1|\,|z_2|,$$

which we prove by means of

$$|z_1z_2|^2 = z_1z_2\overline{z_1z_2} = z_1\overline{z_1}z_2\overline{z_2} = |z_1|^2|z_2|^2 = (|z_1|\,|z_2|)^2.$$

If we use the fact that $|R(z)| \leq |z|$ for any z, property (iii) follows directly from

$$\begin{aligned}
|z_1 + z_2|^2 = (z_1 + z_2)\overline{(z_1 + z_2)} &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\
&= z_1\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_2} + \overline{z_1}z_2 \\
&= |z_1|^2 + |z_2|^2 + (z_1\overline{z_2} + \overline{z_1\overline{z_2}}) \\
&= |z_1|^2 + |z_2|^2 + 2R(z_1\overline{z_2}) \\
&\leq |z_1|^2 + |z_2|^2 + 2|z_1\overline{z_2}| \\
&= |z_1|^2 + |z_2|^2 + 2|z_1|\,|\overline{z_2}| \\
&= |z_1|^2 + |z_2|^2 + 2|z_1|\,|z_2| \\
&= (|z_1| + |z_2|)^2.
\end{aligned}$$

Whenever the complex plane C is mentioned as a metric space, its metric is always assumed to be the usual metric defined above.

The remaining examples to be given in this section fit a common pattern, which we have tried to exhibit in our discussion of Examples 2 and 3. We now point out several major features of this pattern, so that the reader can see clearly how it applies in the slightly more complicated examples that follow.

I. The elements of each space can be added and subtracted in a natural way, and every element has a negative. Each space contains a special element, denoted by 0 and called the *origin*, or *zero element.*

II. In each space there is defined a notion of the distance from an arbitrary element to the origin, that is, a notion of the "size" of an arbitrary element. The size of an element x is a real number denoted below by $\|x\|$ and called its *norm.* Our use of the double vertical bars is intended to emphasize that the norm is a generalization of the absolute value functions in Examples 2 and 3, in the sense that it satisfies the following three conditions: (i) $\|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = 0$; (ii) $\|-x\| = \|x\|$; (iii) $\|x + y\| \leq \|x\| + \|y\|$.

III. Finally, each metric arises as the norm of the difference between two elements: $d(x,y) = \|x - y\|$. As in Example 2, the fact that this is a metric follows from the properties of the norm listed in II. This metric is called the metric *induced by* the norm.

The knowledgeable reader will see at once that we are describing here (though incompletely and imprecisely) the concept of a *normed linear space.* Most of the metric spaces of major importance in analysis are of this type.

Example 4. Let f be a real function defined on the closed unit interval $[0,1]$. We say that f is a *bounded function* if there is a real number K such that $|f(x)| \leq K$ for every $x \in [0,1]$. This concept is familiar to the reader from elementary analysis, as is that of the continuity of f as defined in the introduction to this chapter. The underlying set of points in this example is the set of all bounded continuous real functions defined on the closed unit interval. Actually, the boundedness of such a function is a consequence of its other properties, but at this stage we assume it explicitly. If f and g are two such functions, we add and subtract them, and form negatives, pointwise:

$$\begin{aligned}(f + g)(x) &= f(x) + g(x);\\ (f - g)(x) &= f(x) - g(x);\\ (-f)(x) &= -f(x).\end{aligned}$$

The origin (denoted by 0) is the constant function which is identically zero:

$$0(x) = 0$$

for all $x \in [0,1]$. We define the norm of a function f by

$$\|f\| = \int_0^1 |f(x)|\, dx,$$

and the induced metric by

$$d(f,g) = \|f - g\| = \int_0^1 |f(x) - g(x)|\, dx.$$

The integral involved in this definition is the Riemann integral of elementary calculus. Properties (i) and (ii) of the norm are easy to prove, and (iii) follows from

$$\begin{aligned}\|f + g\| &= \int_0^1 |f(x) + g(x)|\, dx \leq \int_0^1 (|f(x)| + |g(x)|)\, dx\\ &= \int_0^1 |f(x)|\, dx + \int_0^1 |g(x)|\, dx\\ &= \|f\| + \|g\|.\end{aligned}$$

Example 5. The set of points in the preceding example—that is, the set of all bounded continuous real functions defined on the closed unit interval—has another metric which is far more important for our purposes. It is defined by means of

$$\|f\| = \sup \{|f(x)| : x \in [0,1]\},$$

which we usually write more briefly as

$$\|f\| = \sup |f(x)|,$$

and

$$d(f,g) = \|f - g\| = \sup |f(x) - g(x)|.$$

Properties (i) and (ii) of the norm are obvious, and in Problem 5 we ask the reader to prove (iii) in a slightly more general form. This example is typical of a large class of metric spaces which will play a major role in all our work throughout the rest of this book. We denote this space by $\mathcal{C}[0,1]$.

So much for the present for specific examples. We now turn to several fundamental principles relating to metric spaces in general.

Let X be a metric space with metric d. Let Y be an arbitrary non-empty subset of X. If the function d is considered to be defined only for points in Y, then (Y,d) is evidently itself a metric space. Y, with d restricted in this way, is called a *subspace* of X. This technique of forming subspaces of a given metric space enables us to obtain an infinity of further examples from the handful described above. For instance, the closed unit interval $[0,1]$ is a subspace of the real line, as is the set consisting of all the rational points; and the unit circle, the closed unit disc, and the open unit disc are subspaces of the complex plane. Also, the real line itself is a subspace of the complex plane.

It is desirable at this stage to introduce the *extended real number system*, by which we mean the ordinary real number system R with the symbols

$$-\infty \qquad \text{and} \qquad +\infty$$

adjoined. An *extended real number* is thus a real number or one of these symbols. We say (by definition) that

$$-\infty < +\infty;$$

also, if x is any real number, then

$$-\infty < x < +\infty.$$

The symbols $-\infty$ and $+\infty$ add nothing to our understanding of the real numbers. They are used mainly as a notational convenience, as we see below.

Let A be a non-empty set of real numbers which has an upper bound. In Sec. 8 we defined what is meant by the least upper bound (or supremum) of A: $\sup A$ is the smallest upper bound of A, that is, it is the smallest real number y such that $a \leq y$ for every a in A. With the stated assumptions about A, $\sup A$ always exists and is a real number. If A is a non-empty set of real numbers which has no upper bound, and therefore no least upper bound in R, we express this by writing

$$\sup A = +\infty;$$

and if A is the empty subset of R, we put

$$\sup A = -\infty,$$

The greatest lower bound (or infimum) of A is defined similarly: if A is non-empty and has a lower bound, inf A is the largest real number x such that $x \leq a$ for every a in A; if A is non-empty and has no lower bound, we put

$$\inf A = -\infty;$$

and if A is empty, we put

$$\inf A = +\infty.$$

These remarks illustrate one advantage of the extended real number system: it enables us to speak of sup A and inf A for subsets A of the real line without any restrictions whatever on the nature of A.

Another advantage of having available the symbols $-\infty$ and $+\infty$ is that they make convenient a reasonable extension of our concept of an interval on the real line. The reader should refer to the definitions given in Sec. 1 of the various kinds of intervals, for these are the definitions whose scope we are now widening. Let a and b be any two real numbers such that $a \leq b$; then the *closed interval* from a to b is the subset of the real line R defined by

$$[a,b] = \{x : a \leq x \leq b\}.$$

This extends our previous notion in that a closed interval may now consist of a single point (if $a = b$). If b is a real number and a is an extended real number such that $a < b$, then the *open-closed interval* from a to b is

$$(a,b] = \{x : a < x \leq b\}.$$

This allows open-closed intervals of the form $(-\infty,b]$. If a is a real number and b is an extended real number such that $a < b$, then the *closed-open interval* from a to b is

$$[a,b) = \{x : a \leq x < b\}.$$

This permits $[a,+\infty)$ to be considered a closed-open interval. If a and b are extended real numbers such that $a < b$, then the *open interval* from a to b is

$$(a,b) = \{x : a < x < b\}.$$

This adds to the previously defined open intervals those of the form $(-\infty,b)$ where b is real, $(a,+\infty)$ where a is real, and $(-\infty,+\infty)$. Throughout the rest of this book, the term *interval* will always signify one of the four types defined in this paragraph. The extended real numbers a and b are called the *end-points* of these intervals. We have used the symbols $-\infty$ and $+\infty$ with considerable freedom, and it therefore seems desirable to emphasize that an interval in our present sense is always a non-empty subset of the real number system: it never actually contains either of these symbols.

The very definition of a metric space presents us with the concept of the distance from one point to another. We now define the distance from a point to a set and the diameter of a set.

Let X be a metric space with metric d, and let A be a subset of X. If x is a point of X, then the *distance from x to A* is defined by

$$d(x,A) = \inf \{d(x,a) : a \in A\};$$

that is, it is the greatest lower bound of the distances from x to the points of A. The *diameter* of the set A is defined by

$$d(A) = \sup \{d(a_1,a_2) : a_1 \text{ and } a_2 \in A\}.$$

The diameter of A is thus the least upper bound of the distances between pairs of its points. A is said to have *finite diameter* or *infinite diameter* according as $d(A)$ is a real number or $\pm\infty$. We observe that the empty set has infinite diameter, since $d(\emptyset) = -\infty$. A *bounded set* is one whose diameter is finite. A mapping of a non-empty set into a metric space is called a *bounded mapping* if its range is a bounded set. Several of the simpler facts about these concepts are brought out in the following problems.

Problems

1. Let X be a metric space with metric d. Show that d_1, defined by $d_1(x,y) = d(x,y)/[1 + d(x,y)]$, is also a metric on X. Observe that X itself is a bounded set in the metric space (X,d_1).
2. Let X be a non-empty set, and let d be a real function of ordered pairs of elements of X which satisfies the following two conditions: $d(x,y) = 0 \Leftrightarrow x = y$, and $d(x,y) \leq d(x,z) + d(y,z)$. Show that d is a metric on X.
3. Let X be a non-empty set, and let d be a real function of ordered pairs of elements of X which satisfies the following three conditions: $d(x,y) \geq 0$, and $x = y \Rightarrow d(x,y) = 0$; $d(x,y) = d(y,x)$; and $d(x,y) \leq d(x,z) + d(z,y)$. A function d with these properties is called a *pseudo-metric* on X. A metric is obviously a pseudo-metric. Give an example of a pseudo-metric which is not a metric. Let d be a pseudo-metric on X, define a relation $\sim$ in X by means of

   $$x \sim y \Leftrightarrow d(x,y) = 0,$$

   and show that this is an equivalence relation whose corresponding class of equivalence sets can be made into a metric space in a natural way.
4. Let $X_1, X_2, \ldots, X_n$ be a finite class of metric spaces with metrics $d_1, d_2, \ldots, d_n$. Show that each of the functions d and $\bar{d}$ defined